## CONTACT INFORMATION

# JACKET DEMBY'S


**Research Interests**

- Machine Learning
- Deep Learning
- Robotics

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## EDUCATION

**Doctor of Philosophy (Ph.D.)**
Still attending
GPA: 3.724/4

University of Missouri, Columbia, Missouri, USA
**Electrical and Computer Engineering**
Expected Graduation: December 2023

**Master of Science (MS)**
Graduated July 2020
GPA: 3.696/4

University of Missouri, Columbia, Missouri, USA
**Computer Engineering**

**Engineering Diploma**
Graduated July 2015
GPA: 15.06/20

Ucac-Icam Institute, Douala, Cameroon
**Generalist Engineering**
Top of promotion

## RESEARCH EXPERIENCE

**May 2021 – Present: Research Assistant (Vision Guided and Intelligent Robotics Lab – ViGIR Lab)**

- **Behavior-based mobile robotics using multiple sensors (RGB, Intelsense, Sonars, Velodyne)**

**January 2020 – Present: Research Assistant (Vision Guided and Intelligent Robotics Lab – ViGIR Lab)**

- **A study on the approximate solutions provided by data-driven and numerical inverse kinematics solvers for serial robotic manipulators.**
- **Choosing the correct generalized inverse for the numerical solution of the inverse kinematics of incommensurate robotic manipulators.**

**July 2019 – Present: Graduate Student (Vision Guided and Intelligent Robotics Lab – ViGIR Lab)**

- **Object detection and pose estimation using CNN in embedded hardware for assistive technology:** investigating object detection and pose estimation based on stereo vision and deep learning algorithms to be used on resource-limited embedded devices for affordable assistive technology.

**December 2018 – Present: Graduate Student (Vision Guided and Intelligent Robotics Lab – ViGIR Lab)**

- **Inverse kinematics of serial robotics arms using soft-computing methods:** investigating the inverse kinematics results' accuracies provided by two soft-computing methods: artificial neural network and fuzzy neural network.

**August 2017 – December 2018: Graduate Student (Vision Guided and Intelligent Robotics Lab)**

- **Inverse kinematics of serial robotics arms using analytical and hybrid methods:** investigating the effects of the unit choices for robot joints in the computation of the inverse kinematics using analytical and hybrid methods.

## TEACHING EXPERIENCE

**Teaching Assistant,** University of Missouri-Columbia, Columbia, Missouri, United-States

- **Spring 2020** **: Neural Networks**
- **Fall 2020 / 2021 / 2022** **: Building Intelligent Vision-Guided Robots**
- **Spring 2021** **: Introduction to Machine Learning and Pattern Recognition**
- **Spring 2022** **: Power Electronics I**

## TECHNOLOGIES

**Operating Systems**: Linux, Windows, MacOS

**Programming languages**: Python, C/C++, MATLAB, Arduino, Processing, Cuda, Lyx/Latex

**Software and libraries**:

- **Robotics**: ROS (Robot Operating System)
- **Machine learning**: TensorFlow, Keras, Pytorch
- **Computer vision**: OpenCV, PCL
- **Mechanical**: Solidworks, Ansys Fluent

## PUBLICATIONS

1. **J. Demby's**, Y. Gao and G. N. DeSouza, "A Study on Solving the Inverse Kinematics of Serial Robots using Artificial Neural Network and Fuzzy Neural Network," *in 2019 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE 2019),* New Orleans, LA, USA, 2019, pp. 1-6. ***(Accepted)***.
2. **J. Demby's**, Y. Gao, A. Shafiekhani, G.N. DeSouza, "Object Detection and Pose Estimation Using CNN in Embedded Hardware for Assistive Technology, " *in 2019 IEEE Symposium Series on Computational Intelligence (IEEE SSCI 2019)*, December 2019. ***(Accepted)***.
3. **J. Demby's**, A. Shafiekhani, F.B Fritschi and G. N. DeSouza, "Spatio-Temporal Reconstruction and Visualization of Plant Growth for Phenotyping," *in 2021 IEEE Symposium Series on Computational Intelligence (IEEE SSCI 2021)*, December 2021. ***(Accepted)***.
4. **J. Demby's**, J. Uhlmann and G. N. DeSouza, "Achieving Unit-Consistent Pseudo-Inverse-based Path-Planning for Redundant Incommensurate Robotic Manipulators," *in 2023 IEEE International Conference on Robotics and Automation (IEEE ICRA 2023)*. ***(Submitted in September 2022)***.
5. **J. Demby's**, J. Uhlmann and G. N. DeSouza, "Choosing the Correct Generalized Inverse for the Numerical Solution of the Inverse Kinematics of Incommensurate Robotic Manipulators", *IEEE Transactions on Systems, Man, and Cybernetics.* (***In preparation for resubmission***)

## ACHIEVEMENTS AND ACTIVITIES

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|---|---|
| May 2019 – present | Student Member of IEEE. |
| December 2017 | Best 3rd final project in Digital Image Processing class (Fall 2017). |
| August 2017 – present | Member of the University of Missouri Fulbright Organization (**MUFO**). |
| July 2017 – present | Recipient of the **Fulbright Scholarship** in Congo Brazzaville to pursue a M.Sc. in Computer Engineering at the University of Missouri-Columbia. |
| January 2017 | Participant for the Business and Entrepreneurship Track of the Young African Leaders Initiative (YALI Program) in Nairobi, Kenya. |
| April 2016 | 2nd prize ($13,000) of the 2015-2016 national Startup Challenge of Total E&P in Congo Brazzaville. |

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|---|---|
| January 2015 – June 2015 | Participant of the internal Icam exchange program to conduct an industrial mission on Computational Fluid Dynamics (CFD) in Lille, France. |
| June 2012 – July 2013 | Recipient of the **Sogea-Satom scholarship** to study at Ucac-Icam Cameroon. |
| August 2010 – June 2012 | Recipient of the **Total E&P scholarship** to study at Ucac-Icam Congo. |

## SOME COURSEWORK

- Introduction to Mechatronic and Robotic Vision
- Probability and Stochastic Processes for Engineers
- Introduction to Machine Learning and Pattern Recognition
- Parallel Programming for High Performance Computing
- Introduction to Computational Intelligence
- Digital Image Processing
- Computer Vision
- Supervised Learning
- Neural Networks and Deep Learning
- Real-Time Embedded Systems

## SOME COURSEWORK PROJECTS INVOLVING FINAL PRESENTATIONS

**Class: Real-Time Embedded Systems (Spring 2020)**
**Project**: Data generation for supervisory control and data acquisition system (DataGen-SCADA)
**Main objective, results and tools**: Build a real-time data generation simulator for distributed SCADA systems using IoT devices; to gather and analyze data in real-time industrial settings and behaviors. Python, Processing, Arduino and ESP8266 devices were used.

**Class: Neural Networks (Spring 2019)**
**Project**: Solving the inverse kinematics of a 4 DoF SCARA robot using extreme learning machines and radial basis function networks
**Main objective, results and tools**: Conduct a comparative study between extreme learning machines and radial basis function networks in solving the inverse kinematics of the chosen robotic arm. Lowest MSE (Mean-Squared Error) results were obtained with extreme learning machines. Matlab and Python were used.

**Class: Introduction to Computational Intelligence (Fall 2018)**
**Project**: Solving the inverse kinematics of a 6 DoF Kinova robot using artificial and fuzzy neural networks
**Main objective, results and tools**: Conduct a comparative study between artificial and fuzzy neural networks in solving the inverse kinematics of the chosen robotic arm. A trade-off was observed in terms of precision of the predicted joint configuration depending on the workspace. Matlab and Python were used.

**Class: Computer Vision (Spring 2018)**
**Project**: Hardware integration of a real time object detection CNN (Yolo V2)
**Main objective, results and tools**: Develop an android mobile application based on YOLO V2 to recognize and pronounce objects in real-time. Android, Darknet and a pretrained Yolo V2 model were used.

**Class: Introduction to Mechatronic and Robotic Vision (Fall 2017)**
**Project**: Investigating the control of a robotic arm with a real-time tracking of human hand motions
**Main objective, results and tools**: Track human hand motions using a Kinect sensor and control the robotic arm with the hand motions tracking information. C/C++ and ROS were used.

**Class: Digital Image Processing (Fall 2017)**
**Project**: Super Image Restoration (SIR)
**Main objective, results and tools**: Develop a Matlab application with a friendly Graphical User Interface (GUI) to restore old damaged images with cracks in only three click steps. Matlab, Python and Algorithmia Colorization API were used.

## Most Recent Work Experience

**Junior Project Engineer** | December 2016 – July 2017
**Total E&P Congo - Pointe-Noire, Republic of Congo**

- Prepared, supervised and generated daily reports for construction activities of the Floating Production Unit (FPU) Likouf alongside contracted companies.
- Worked on Pressure Safety Valves (PSV) Recalibration Campaign, managed a database of the PSV joint stocks, and ensured team members compliance with on-site safety measures.

**Maintenance Engineer** | September 2015 – October 2016
**Sogea-Satom - Brazzaville, Republic of Congo**

- Monitored the computer-assisted supply chain and maintenance management software and produced monthly analysis reports.
- Developed and maintained Excel VBA (Visual Basic for Applications) programs for monthly stocks' management analysis, optimization and periodic inventories follow up.

**Intern Maintenance Engineer by Alternance** | January 2013 – December 2014
**Sogea-Satom - Brazzaville, Republic of Congo**

- Monitored the computer-assisted supply chain and maintenance management software.
- Continually improved methods and procedures for processes, stocks' management, documenting and workflow techniques.

**Intern Maintenance Technician** | January 2012 – March 2012
**Total E&P Congo – Pointe-Noire, Republic of Congo**

- Determine the causes of cascade regulations of three-phase separators of Katina treatment unit at Djeno Oil Terminal and propose improvement solutions to ameliorate their control/regulation cycles.
- Conduct a study on the appropriate type of sensors (ultrasonic sensors to detect water, gas and oil levels) to be replaced on those separators and suggest vendors.

## Languages and Interests

Languages

- French: native language
- English: fluent (speaking, reading, writing)

Interests

**Sport**: Walking, Running, Volleyball, Swimming
**Board games**: Scrabble, Ludo, Chess, Checkers

## Volunteer Work

**Intervarsity at University of Missouri-Columbia** | Volunteer photographer and events' organizer

- Organize events (games, discoveries, English clubs, shopping, etc.) to help international students feel welcome at the university.
- Take pictures during the events and share them with all the members.

**University of Missouri Fulbright Organization (MUFO) |** Volunteer

- Work with the International Center to provide rides to new international sponsored students.
- Lead scavenger hunt activities to help international sponsored students discover and locate key facilities and resources on campus.

**OVA (Œuvre de Vie et d'Amour) Association in Cameroon** | Volunteer

- Organize activities on the International Day of the African Child and Christmas for street children and children with special needs.